

Research Project

Title: *Multilingual Misinformation and Narrative Detection: Transformer- and Retrieval-Augmented Methods under a Responsible-AI Lens* **Area:** National PhD in AI — "AI & Society" (University of Pisa, KDD Lab) **Intended supervision:** Prof. Dino Pedreschi / Prof. Salvatore Ruggieri (*verify specific advisor on call PDF*) **Applicant:** Nauval Zulfikar, MSc Business Analytics (Aston University, 2024, 1:1 First Class)

1. Motivation and problem

Misinformation does not spread as isolated false claims; it spreads as *narratives* — coherent, evolving framings that recur across communities and languages and that detection systems trained on single-language, claim-level data routinely miss. Two gaps are especially acute for the "AI & Society" agenda. First, most misinformation classifiers are evaluated on English and degrade sharply on lower-resource languages, leaving large electorates (e.g. Indonesian-speaking populations) comparatively unprotected. Second, the detectors that *do* work are typically opaque, which is untenable when their outputs feed moderation or electoral-integrity decisions that demand contestability and auditability.

This project asks: **can transformer- and retrieval-augmented methods detect cross-lingual misinformation narratives accurately *and* transparently, in a way that is auditable enough to be deployed responsibly?**

2. Research questions

- **RQ1 (Multilingual detection):** How well do aspect- and stance-aware transformer models transfer across languages for narrative-level misinformation detection, and where does transfer fail?
- **RQ2 (Narrative grounding via RAG):** Can retrieval-augmented generation, grounded in a curated evidence base, improve detection *and* yield traceable justifications rather than free-form claims?
- **RQ3 (Responsible-AI lens):** What governance instruments — bias auditing across languages/communities, explanation faithfulness, and post-deployment drift monitoring — are needed before such a system can be trusted in a civic context?

3. Approach and methods

Foundation in prior work. This builds directly on completed projects. For the **2024 Indonesia Election Aspect-Based Sentiment Analyser** (Indonesia AI), I worked on aspect-based sentiment over electoral discourse — the same modelling primitive that, reframed as

stance and narrative, underpins misinformation detection. My **MSc dissertation** fine-tuned **DeBERTa-v3** for multi-aspect classification on review text and evaluated emergent biases against literature baselines. The **Pfizer Public Sentiment Analysis** project (Qarir Generator), where I led sentiment-analysis development, and the **NLP-based competitor app-review analysis** at Bank Muamalat both exercised production sentiment pipelines on noisy real-world text. The **CV-JD Suitability Checker** and **LLM-Shipper-Profiles** projects gave me a working transformer + RAG stack (DeBERTa-v3, sentence-transformers, an LLM augmented with retrieval over a curated knowledge base, with a traceable rule-based fallback) — the exact architecture this proposal extends.

Workstream A — Multilingual narrative representation (RQ1). Fine-tune transformer encoders for stance/aspect labelling on misinformation narratives, then probe cross-lingual transfer (English → Indonesian and other languages, subject to data availability). Aspect-based framing lets a model say *which* element of a narrative is contested, not merely true/false. Use controlled, literature-anchored evaluation — a methodology I have already practised in a 450-run agent-based parameter sweep — rather than single split-point reporting.

Workstream B — Retrieval-augmented justification (RQ2). Pair the detector with a retrieval layer over a curated evidence base (fact-checks, prior debunks, source provenance) so that a flag is accompanied by retrieved, inspectable evidence. Retain a transparent fallback when retrieval confidence is low — the reproducible-and-ablatable design principle I applied in LLM-Shipper-Profiles — so the system fails *legibly*.

Workstream C — Responsible-AI evaluation (RQ3). Measure disparate error rates across languages and communities; assess explanation faithfulness (do retrieved justifications reflect the model's actual decision?); and specify drift/staleness monitoring for deployment. This draws on my Bank Muamalat experience maintaining model reliability under regulatory and audit frameworks — validation cycles, staleness monitoring, audit-grade documentation — i.e. the *technical preconditions* of responsible AI, treated here as first-class research objects rather than afterthoughts.

4. Indicative three-year plan

- **Year 1:** Literature synthesis (misinformation, computational social science, multilingual NLP, XAI); assemble/curate multilingual narrative datasets; reproduce strong baselines; establish the evaluation harness.
- **Year 2:** Develop the multilingual narrative detector (A) and the RAG justification layer (B); run cross-lingual transfer and ablation studies; first publication.
- **Year 3:** Responsible-AI evaluation (C) — bias auditing, explanation faithfulness, drift monitoring; consolidate into a deployable, auditable pipeline; thesis and dissemination.

5. Fit with the "AI & Society" area

The proposal sits squarely on the area's stated themes — misinformation, disinformation, social impact of AI, responsible AI, computational social science, explainable AI — and on KDD Lab's tradition of explainable and socially-aware data mining. My contribution is a candidate who has *already* built transformer + RAG misinformation-adjacent systems on real electoral and public-health text, and who has operated ML under audit/governance constraints. The methodological notes (datasets, baselines, exact metrics) are deliberately framed to be refined with the supervisor against the official research-project template.

No results, citations, or affiliations beyond the applicant's verified portfolio are claimed. Specific datasets, baselines, and supervisor are to be finalised against the official XLII call and in discussion with KDD Lab. [TODO: verify with user] where the call PDF dictates section/length requirements.